

Practice Problems 7

ECON 441: Introduction to Mathematical Economics

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Exercise 5.3

1. Use the determinant $\begin{vmatrix} 4 & 0 & -1 \\ 2 & 1 & -7 \\ 3 & 3 & 9 \end{vmatrix}$ to verify the first four properties of determinants.
4. Show that when all the elements of an n th-order determinant $|A|$ are multiplied by a number k , the result will be $k^n|A|$.
5. Calculate the determinant for the following matrices. Comment on whether the matrices are nonsingular and the rank of each matrix.

a) $\begin{bmatrix} 4 & 0 & 1 \\ 19 & 1 & -3 \\ 7 & 1 & 0 \end{bmatrix}$

b) $\begin{bmatrix} 4 & -2 & 1 \\ -5 & 6 & 0 \\ 7 & 0 & 3 \end{bmatrix}$

c) $\begin{bmatrix} 7 & -1 & 0 \\ 1 & 1 & 4 \\ 13 & -3 & -4 \end{bmatrix}$

d) $\begin{bmatrix} -4 & 9 & 5 \\ 3 & 0 & 1 \\ 10 & 8 & 6 \end{bmatrix}$

8. Comment on the validity of the following statements:
 - (a) Given any matrix A , we can always derive from it a transpose and a determinant.
 - (b) Multiplying each element of an $n \times n$ determinant by 2 will double the value of that determinant.
 - (c) If a square matrix A vanishes, then we can be sure that the equation system $Ax = d$ is nonsingular.

Exercise 5.4

2. Find the inverse of each of the following matrices:

a) $A = \begin{bmatrix} 5 & 2 \\ 0 & 1 \end{bmatrix}$

b) $B = \begin{bmatrix} -1 & 0 \\ 9 & 2 \end{bmatrix}$

c) $C = \begin{bmatrix} 3 & 7 \\ 3 & -1 \end{bmatrix}$

d) $D = \begin{bmatrix} 7 & 6 \\ 0 & 3 \end{bmatrix}$

3. (a) Drawing on your answers to Prob. 2, formulate a two-step rule for finding the adjoint of a given 2×2 matrix A : In the first step, indicate what should be done to the two diagonal elements of A in order to get the diagonal elements of $\text{adj}A$; in the second step, indicate what should be done to the two off-diagonal elements of A . (Warning: This rule applies only to 2×2 matrices.)
- (b) Add a third step which, in conjunction with the previous two steps, yields the 2×2 inverse matrix A^{-1} .

4. Find the inverse of each of the following matrices:

$$\text{a) } E = \begin{bmatrix} 4 & -2 & 1 \\ 7 & 3 & 0 \\ 2 & 0 & 1 \end{bmatrix}$$

$$\text{b) } F = \begin{bmatrix} 1 & -1 & 2 \\ 1 & 0 & 3 \\ 4 & 0 & 2 \end{bmatrix}$$

$$\text{c) } G = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 0 & 1 \\ 0 & 1 & 0 \end{bmatrix}$$

$$\text{d) } H = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}$$

6. Solve the system $Ax = d$ by matrix inversion, where

$$\text{a) } 4x + 3y = 28$$

$$2x + 5y = 42$$

$$\text{b) } 4x_1 + x_2 - 5x_3 = 8$$

$$-2x_1 + 3x_2 + x_3 = 12$$

$$3x_1 - x_2 + 4x_3 = 5$$

7. Is it possible for a matrix to be its own inverse?

Exercise 5.5

1. Use Cramer's rule to solve the following equation systems:

$$\text{a) } 3x_1 - 2x_2 = 6$$

$$2x_1 + x_2 = 11$$

$$\text{c) } 8x_1 - 7x_2 = 9$$

$$x_1 + x_2 = 3$$

$$\text{b) } -x_1 + 3x_2 = -3$$

$$4x_1 - x_2 = 12$$

$$\text{d) } 5x_1 + 9x_2 = 14$$

$$7x_1 - 3x_2 = 4$$

2. For each of the equation systems in Prob. 1, find the inverse of the coefficient matrix, and get the solution by the formula $x^* = A^{-1}d$.
3. Use Cramer's rule to solve the following equation systems:

$$(a) 8x_1 - x_2 = 16$$

$$2x_2 + 5x_3 = 5$$

$$2x_1 + 3x_3 = 7$$

$$(d) -x + y + z = a$$

$$x - y + z = b$$

$$x + y - z = c$$